Cross-fitted causal erasure separates subject and attractor number in DistilGPT-2

Agreement attraction places two number values in competition. In (1), the hierarchical subject is singular and the closest noun is plural. This contrast also exposes a familiar problem in mechanistic interpretation: a probe can recover a feature that downstream layers ignore (Elazar et al. 2021), and a specialized attention head may have little control over the output (Timkey & Linzen 2026). We test which number representation contributes to the next-word preference of DistilGPT-2. Nearest-noun number reaches perfect probe accuracy, while cross-fitted erasure of a subject-correlated direction produces a 1.578-logit loss relative to attractor-direction erasure. The effect appears late and survives a lexical-cluster bootstrap.

(1) *The key near the cabinets {is/are}...*

Question. Causal work has localized agreement information to model components and residual-stream directions (Finlayson et al. 2021; Ferrando & Costa-jussà 2024). GPT-2 also shows attraction effects compatible with similarity-based interference (Ryu & Lewis 2021). These findings leave open whether the accessible representation of the nearby noun contributes to inflection. A nearest-noun account predicts a substantial effect when its fitted direction is removed, with the largest cost on conflicting-number items. A hierarchical account predicts a larger cost for the fitted subject direction.

Materials and behavioral result. We generated 80 prefixes from ten lexical sets. The design crossed subject and attractor number within two structures. In the PP condition, the attractor was inside a subject-modifying PP as in (1); in the relative-clause condition it was the embedded subject, as in *The key that the cabinets can describe*. All target forms, *is* and *are*, are single tokens. We measured their next-token logit contrast in DistilGPT-2 (81.9M parameters). The model selected the grammatical form on 92.5% of items (74/80). Accuracy was 97.5% for number-congruent items but 87.5% for incongruent items; it was 100% in PPs and 85% in relative clauses. The six errors were concentrated in relative clauses, especially those with plural subjects (75% accuracy). We therefore treat the analysis as a controlled pilot. Its scope does not extend to agreement behavior across the model family.

Diagnostic analysis. At the final prefix position, we extracted the embedding state and the residual stream after each of six Transformer blocks. Separate logistic probes predicted subject and attractor number. Each of five folds held out two complete lexical sets, including both structures and every number condition. Subject-number accuracy is at chance in the embedding state and reaches 93.75% after block 5. Attractor number is easier to decode and reaches 100% after blocks 3 and 4.

Table 1: Group-cross-validated probe accuracy (%) and cross-fitted loss in the correct-inflection logit margin. State 0 is the embedding output.

State	Subj. probe	Attr. probe	Subj. erase	Attr. erase	Random
0	50.00	75.00	–	–	–
1	71.25	86.25	0.003	0.001	0.000
2	58.75	97.50	0.008	−0.006	0.002
3	73.75	100.00	0.019	−0.030	−0.001
4	91.25	100.00	0.332	−0.005	0.000
5	93.75	97.50	0.712	−0.003	−0.000
6	92.50	88.75	1.578	0.000	0.003

Causal intervention. Each test item received a direction and center fit on the other eight lexical sets. We separately projected its final-position state away from the fitted subject and attractor directions, then ran the remaining blocks. Ten random directions per fold supplied a dimensionality-matched control. The subject-direction loss is small through block 3, then grows to 0.332, 0.712, and 1.578 logits after blocks 4–6. Final accuracy becomes 66.25%, against a 92.5% baseline; incongruent accuracy becomes 55%, against 87.5%. The fitted attractor direction changes the margin by less than 0.001 logits and leaves both accuracies unchanged. Random-direction erasure changes the final margin by 0.003 logits. The final subject-minus-attractor contrast is 1.578 logits; a 10,000-sample bootstrap over lexical sets gives a 95% CI of [1.460, 1.686].

Analysis. The residual stream preserves linearly recoverable information about the controller and the nearby distractor. Their fitted directions have different causal effects. The subject-correlated direction contributes to the model's late logit contrast even at states where attractor number has higher probe accuracy. Probe accessibility therefore overstates the causal role of the fitted attractor direction.

The late onset agrees with Ferrando and Costa-jussà's (2024) subject-number direction in the final residual stream and with Finlayson et al.'s (2021) structure-dependent agreement mechanisms. The evidence supports a narrow claim: in this pilot, a cross-fitted subject-correlated direction contributes to the copular logit contrast. Under the same intervention, the most accessible linear attractor direction has no measurable effect. The erasure effect is similar across PPs and relative clauses (1.616 versus 1.540 logits), though it is asymmetric by subject number. Final erasure lowers singular accuracy to 42.5%; plural accuracy remains at 90%. The fitted axis may combine number with a singular-specific output bias. A one-dimensional null effect also leaves other attractor representations untested. Replication should use larger stimulus sets and model families, then compare projection methods such as LEACE or mean projection (Dobrzeniecka et al. 2025).

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